

Predicting Outcomes of NBA Games

Mian Abdullah Zahid
Department of Computer Science
Western University
London, ON
mzahid33@uwo.ca

Bilal Naeemuddin
Department of Science
Western University
London, ON
bnaeemud@uwo.ca

Waleed Chaudhry
Department of Science
Western University
London, ON
wchaudh4@uwo.ca

Abstract—Predicting the outcome of NBA games remains a difficult task due to factors such as player performance and injuries. This project builds multiple machine learning models with features from the Kaggle Repository named NBA Datasets. This repository contains multiple datasets including: player performance metrics, league rankings, game details, as well as IDs that help join the datasets together which can be used for feature extraction and model building. The data is split chronologically, preventing any occurrence of time leakage and providing unbiased results on unseen future data. The resulting models used a 10 game rolling window of the previous results for the respective matchup being predicted and achieved the following when tested on league-wide data: Random Forest Classifier achieved an accuracy of 69.6% and an ROC-AUC of 77.9%, Logistic Regression achieved an accuracy of 70.7% and ROC-AUC of 78.6%, and XGBoost achieved an accuracy of 71.3% and ROC-AUC of 79.1%. The evaluation metrics saw an increase when testing the model on individual team seasons such as the Toronto Raptors 2018-2019 season where they won the NBA Championship. The Random Forest Classifier achieved an accuracy of 82.0% and ROC-AUC of 91.9%, the Logistic Regression Model achieved an accuracy of 69.7% and ROC-AUC of 80.8%, and the XGBoost Model achieved an accuracy of 70.8% and ROC-AUC of 81.1%. The Raptors case study enforces the notion that team statistics and player momentum are important factors that are often overlooked by models that are generalized. Future work will include adding real-time data to enhance the models predictive accuracy and capture in-game variance better, and fine-tuning the models accordingly for better results. A key takeaway was that the rolling 10 game window and cumulative team and player statistics improve predictive power and performance compared to models only using seasonal aggregate statistics.

I. INTRODUCTION

Many studies rely on static features, neglecting how dynamic player performance and contextual factors (e.g home-court advantage) affect the outcomes. The drastic rise in popularity of sports analytics has transformed the sports industry influencing factors such as coaching strategies, roster moves, scouting reports, and even betting markets. Despite these analytical advances, accurately predicting the outcome of a basketball game remains a significant challenge due to in-game factors that are often overlooked in current predictive models, such as player injuries and game momentum. Current approaches to this problem combine in-game team box-score statistics such as points, rebounds, and assists without accounting for much historical data. Another problem faced is that these models ignore how recent historical games,

player performance (player points, rebounds, assists, etc), and contextual factors (home-court advantage) could impact the predictive results of games.

This project aims to investigate this problem and design multiple machine learning models capable of accurately predicting if the home team will win a game using the previous 10 games of that match-up as recent historical data to predict accurate home-team wins. The main objective is to select the optimal machine learning model that will give the best performance in predicting NBA games. We selected three predictive models, Logistic Regression, Random Forest, and XGBoost, because these models are supervised models, meaning that they learn patterns from historical data to predict outcomes of NBA games while balancing interpretability and predictive power.

We choose to use a collection of CSV files found on Kaggle that contains datasets that include historical player data (Individual per-game statistics), historical team-based statistics (Box Score Statistics), and NBA games from the 2004 to 2022 seasons including pre-season, regular-season, and playoff games. This will ensure that the model will be trained to learn how to predict each game in a real environment as it will not be able to use statistics from the current game predicted, relying on the past 10 games of the specific matchup preventing any look-ahead bias.

II. BACKGROUND AND RELATED WORK

In this section, we will explore the existing work in NBA game prediction, focusing on machine learning. We will discuss the trends and challenges in prediction, and identify the gaps that this paper attempts to fill.

A. NBA Prediction and Machine Learning

Machine learning (ML) in sports analytics has gained prominence in determining the result of a game, especially in high-velocity sports like basketball. Sports analytics has been entirely transformed over the past decade due to the shift of simple statistical models of sports to more complex ML algorithms, such as ensemble models and deep learning algorithms, which in turn can reflect non-linear relationships between features and increase the accuracy of prediction [1]. Earlier techniques, such as logistic regression, were partially effective as they failed to address the complex pattern of features, which may influence a game outcome. Studies show

non-linear models like Random Forest and deep learning techniques provide significant improvement over these traditional models [2] [3]. These approaches struggle with non-linear and complex relationships that exist between variables such as the contribution of individual players and team strategy [4].

This paper is aligned to the evaluation of ML models that are more accurate at predicting by adopting non-linear decision-making processes hence makes it relevant to the application of Random Forest and XGBoost in this article

B. Feature Engineering Trend

Feature engineering has become a major focus in the research. The feature engineering has been gradually being considered important in the world of sports analytics. The research has found that team-level features, including offensive efficiency, defensive efficiency, rebound rates and turnover rates are very predictive of the outcome of game results [3][7]. The addition of these detailed team statistics has proven to work even better than the simple win-loss statistics and resulted in more precise predictions [3][5][6][7]. The paper [9] also found that combining individual players performance with the team level statistics has been proven to be essential in predicting games. Metrics including points per game (PPG), assists per game (APG), and player efficiency ratings (PER) are also commonly used [11]. Recent studies [5][9] have found that cumulative player statistics (i.e., performance over multiple games or seasons) can improve prediction accuracy. The other features from another report [8][11] that stand out include the contextual features, such as home-court advantage, which was always given a 60 percent win probability to home teams, resting days, distance covered, and the injury reports have also been included in the model. Research has indicated that the variables have high effects on the results of games, and are critical in enhancing the performance of the model, especially in high velocity games such as playoffs.

C. Ensemble Methods

The paper [1] explores the limitations of traditional models in sports prediction highlighting how ensemble methods have proven to outperform simpler techniques by combining multiple decision trees for improving generalization, reducing overfitting and improving the overall accuracy [2][5][9]. There have been a recent shift towards ensemble techniques, such as Random Forest and XGBoost, that provide strong ability to model complex and high-dimensional data. The ensemble techniques involve combining together multiple decision trees or models to achieve greater generalization, reduce overfitting and increase prediction accuracy. These methods have been popular with sports analytics to predict games, because they are capable of non-linearities and complex feature interactions compared to traditional ones [3][5][9][10]. The paper [5] reported that the accuracy of Random Forest is higher than traditional algorithms, such as the Logistic Regression, with a random forest achieving an accuracy of 83.78% against the accuracy of the Logistic Regression algorithm, 81.64%. Their research emphasizes the higher efficiency of ensemble

techniques in game results forecasting, which makes Random Forest a good candidate to be compared in our work.

D. Research Gaps

Despite the progress in the field of NBA prediction studies, there are still a number of gaps. Many models treat historical data equally, ignoring how recent team form and player stats impact game outcomes. The papers identified the limitation that the static approach fails to capture the evolving nature of team performance, and our research aims to address fill this gap by using cumulative statistics and recent performance trend to reflect the seasonal dynamics.

Model interpretability is another gap. Although both the Random Forest and the XGBoost provide good accuracy but these models require better transparency. In order to enhance the interpretability, our study focuses on feature importance scores to ensure that the models are not only effective but can additionally be used to provide actionable insights.

Furthermore, many studies rely on static features, neglecting how dynamic player performance and contextual factors (e.g home-court advantage) affect the outcomes. The research papers we examined demonstrated that these factors improve the accuracy of the model. Our research will incorporate dynamic features like rolling averages and cumulative stats following recent trends in contextual data in sports analytics.. Another constraint in the research is the lack of standardized benchmarking, with models being generally evaluated on different datasets or sets of features, so the outcomes can not be easily compared across studies.

III. METHODS

A. Research Objectives

O1: To build a machine learning model to predict outcomes in the NBA games based on the set of extensive features.

Significance: The purpose will be to develop a tool that can be applied by sports analysts, coaches, and even fans to determine whether the home team will win or lose. This will also help in decision-making, strategic planning and even in betting markets, it would give a more accurate insight based on data-driven predictions.

O2: To evaluate the performance of the different machine learning algorithms namely Logistic Regression (LR), Random Forest (RF), and XGBoost.

Significance: This study will determine the most appropriate algorithms to apply in the tasks of making predictions in the NBA that can assist in addressing the question of the practicality of the algorithms in sports analytics in real-life scenarios and enhance the quality of game predictions.

O3: Incorporate the dynamic attributes like rolling averages and cumulative statistics to determine their effects on the predictive power of the model.

Significance: This will demonstrate the necessity to consider the idea of temporal dynamics in sports predictions, which will enhance the accuracy of models by responding to the alterations in the team performance and player form in real-time.

O4: Evaluate the interpretability of the XGBoost and the Random Forest models using the scores of the feature importance, and compare them to Logistic Regression.

Significance: This ensures while the models are highly accurate, they are also transparent, giving the stakeholders a chance to know the logic generating the predictions and make the models more practical to be utilized in practice.

B. Research Methodology

This section discusses the methodology we used for building our predictive model

1) *Data Collection and Preprocessing*: The dataset used in this study is available publicly on the web at Kaggle offering extensive historical information on NBA games. The data is organized into several CSV-files:

games.csv: This table contains records of every game with points played in it, win/loss index, team ID, and date.

gamesdetails.csv: Includes details of per-game player statistics (points, rebounds, assist, etc.).

players.csv: This contains information about players, including player name, team ID and other identifiers.

ranking.csv: This data has seasonal rankings of the teams, win percentages, and conference data.

teams.csv: Has metadata of teams, e.g., the arena capacity, year of formation, etc.

We merged these data sets to generate a set of features that describe macro-level trends in team performance along with micro-level contributions of players. This helped us to build a comprehensive set of features for game outcomes.

2) *Data Cleaning*: Numeric columns were imputed with the mean and categorical columns with the mode, to deal with missing data. Duplicate entries (e.g. duplicate game or player statistics) were removed to avoid skewing of the results. Team names were normalized within the set of data. Dates in the game were formatted into the same format that could be sorted and filtered easily. This made sure that time division between training and testing of the model was correct.

3) *Data Preprocessing*: In our data preprocessing phase, in order to avoid time leakage, we implemented strict measures to avoid a model cheating by relying on future data to predict previous events. In this regard, time leakage implies the utilization of the future game information (that would not be available at the time of prediction) to forecast the past games.

The measures included: Sort the complete data in terms of game date. Summing up the statistics of a team (e.g., the average points made, the win percentage, etc.) based solely on the games that a team has previously played and the current result. This was achieved through a rolling window of game statistics so that past predictions would not be influenced by future game statistics. Player statistics were also aggregated by leaving out all the games played prior to the actual game. We divide the data in time, i.e., the training set must include the previous games, while the test set includes the last games.

4) *Feature Engineering*: Team-Level Features: Cumulative Win Percentage: The win percentage of the team is computed using the past 10 matches with the above formula. Average

Points Scored and Conceded: This is calculated by averaging (points conceded and points scored) in the last 10 games. Average Rebounds and Assists: Average rebounds and assists per game were computed as well in the past 10 games.

Player-Level Features: Average points per game: The number of points, on average, each of the players has scored across the games. This indicates the potential of the scoring of individual players. Star Player Scoring Power: The highest total number of points earned by one individual over the past 10 games, the star offensive player of the team. Roster Depth: The player's scoring averages of the past 10 games added together. This statistic will help understand the degree of reliance of the team on individual players and/or collective scoring skills.

Contextual Features: Team Ranking (Win Percentage): This is the current team's win percentage in the season, and it is used to gauge the performance of a team. Conference: This can determine the team-to-team matchings and performance dynamics as a team is either in the Eastern or Western conference. Arena Capacity: Home-court advantage is obtained with the capacity of the home arena of the team. The use of home-court advantage was adopted as a significant contextual factor since teams usually play better on their home court because of the familiarity with the arena and fan support.

Home-Court Advantage Feature: A binary variable was developed on home-court advantage, where: 0 1 = Home game (team playing in home arena). 0 0 = Away game (team is playing in the field of the opponent). Team Founding Year: This feature is added as a historical feature, though it had an indirect effect.

C. Model Architecture and Training

For our research, we selected three machine learning models: Logistic Regression (LR), Random Forest (RF), and XGBoost. These models were selected because of their strengths, adaptability, and proven track records in handling predictive tasks involving complex and structured data, and they fit our dataset and the type of project.

1) *Logistic Regression*: The Logistic Regression is a linear model that has been commonly used in the field of binary classification. It is applicable in scenarios where the association between the input features and the outcome to be obtained is linear or almost linear. Although it might not capture all the aspects of the prediction of the NBA game, it is a good model of baseline.

a) Logistic Regression Configuration: Max Iterations: 1000. This will guarantee that the solver will be able to have enough iterations to find an optimal solution due to the large dimension of the feature set. Regularization: L2 regularization (Ridge) was used in order to reduce overfitting when large coefficients were penalized. Random State: 42 to maintain consistency with other models for comparative analysis. Solver: the default optimization algorithm, 'lbfgs' was selected, since it is applicable to multiclass and binary problems whose datasets are relative large in size.

2) *Random Forest*: Random Forest is an ensemble technique of learning that involves integrating the forecasts of a number of decision trees, where all of the trees are trained in random selections made on the data. This method allows the model to handle non-linear relationships and interactions between features. This model will have the ability to evaluate not only team-level statistics but also individual player contributions. It works very well with missing data, and is less overfitting than single decision trees, so it can be used on real-world data, which may not necessarily be clean or complete.

a) *Random Forest Configuration*: Number of Trees: 300 trees were used to achieve stability and strength in the model. Balanced Class Weights: This was done to eliminate the issue of class imbalance because the home teams tend to win more. Random State: 42 to obtain reproducibility. We performed hyperparameter tuning using a grid search to optimize the number of trees in the Random Forest and other relevant parameters. However, the default configuration of 300 trees with balanced class weights performed effectively for this task.

3) *XGBoost*: XGBoost (Extreme Gradient Boosting) always works well compared to many other models in classification problems, particularly structured/tabular data such as NBA statistics. It uses boosting, in which the decision trees are constructed one after another and each new tree tries to rectify the errors of the previous one. Since NBA game results may be affected by minute details in the performance of the teams and the context of the game, XGBoost has an advantage of more effectively learning these complexities. It is a great choice to reduce the impact of overfitting by regularization of the model and, therefore, be able to predict the model on unknown data.

a) *XGBoost Configuration*: N-Estimators: 100. This is the number of boosting rounds or sequential trees obtained so as to minimize the loss function. Learning Rate (eta): 0.1. To ensure that the boosting process will not overfit the best solution, a smaller learning rate was selected. Max Depth: 3. Limiting the depth of individual trees to avoid noise. Eval Metric: 'logloss'. For minimizing the logistic loss Random State: 42. Consistent as we did for our random forest model

D. Model Implementation and Evaluation Process

1) *Rolling Window Process*: To simulate real-world conditions, the model uses a rolling window of 10 games for calculating key features such as cumulative win percentage and average points scored. We calculated features including win percentage, points scored, rebounds and assists of each team and each player so that no data of the future was used to predict the past.

$$\text{Cumulative Win Percentage} = \frac{\text{Total Wins in Last 10 Game}}{\text{Total Games Played in Last 10 Games}}$$

$$\text{Average Points Scored} = \frac{\sum \text{Points Scored in Last 10 Games}}{10}$$

$$\text{Average Points per Game} = \frac{\sum_{i=1}^{10} \text{Points in Game } i}{10}$$

E. Model Evaluation

1) *Evaluation Metrics*: The performance was evaluated based on the following important metric: Accuracy: The overall percentage of correct predictions. Precision: Measures how many of the predicted positive outcomes (home team wins) were actually correct. Formula for Precision:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Where: TP: True Positives (wins of the home team that are predicted correctly). FP: False Positives (the home team falsely predicted as a winner when the home team was defeated). Recall: Measures how many of the actual positive outcomes (home team wins) were correctly predicted by the model. F1-Score: The harmonic mean of Precision and Recall, providing a balance between the two metrics.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

ROC-AUC: This metric assesses how well the model can distinguish between home team wins and losses.

IV. 6. RESULTS

In this section, we provide a detailed empirical analysis of our three predictive architectures. We break down the results by examining the statistical relationships between features, the classification performance across the entire league, and a specialized case study of an elite championship roster.

Feature Interaction and Correlation Analysis: To justify our feature engineering process, we first analyzed the linear dependencies within our dataset.

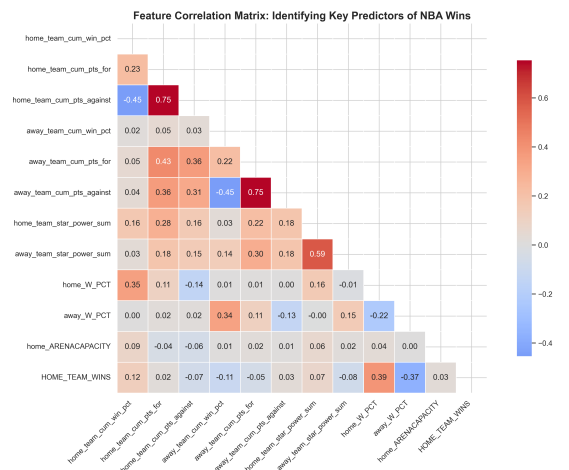


Fig. 1. Feature correlation matrix.

The correlation matrix reveals that a team's standing (W_PCT) and its cumulative season win percentage are the

most significant linear predictors of a game's outcome. Interestingly, we also found even a moderate positive relationship between our own custom measure of Star Power and the final outcome. This confirms that aggregating individual player scoring history provides a unique "talent-gap" signal that is not fully captured by team-level standings alone. By visualizing these relationships, we confirmed that while some features are redundant tree-based models like Random Forest and XGBoost are capable of utilizing these overlapping signals to improve decision-making.

B. Comparative Model Performance: We evaluated the models on an unseen test set of 4,007 games. The results indicate that ensemble methods provide a measurable advantage over linear baselines when dealing with the high variance of professional basketball. **XGBoost Performance:** This model achieved the highest robustness with an Accuracy of 71.25% and an ROC-AUC of 79.09%. The high score proves that the model is excellent at ranking the probability of a win, allowing for better distinction between "close calls" and "certain victories". **Logistic Regression Baseline:** Despite its linear nature, this model remained highly competitive with 70.70% Accuracy and a 78.64% ROC-AUC. This suggests that much of the variance in NBA games can be explained by linear combinations of season-long momentum and roster talent. **Random Forest Metrics:** While recording the lowest general accuracy at 69.60%, the Random Forest model prioritized sensitivity toward winning conditions, as evidenced by its superior recall metrics.

C. Error Distribution and Confusion Matrix Analysis: To validate the models practical utility, we examined how they categorized wins and losses.

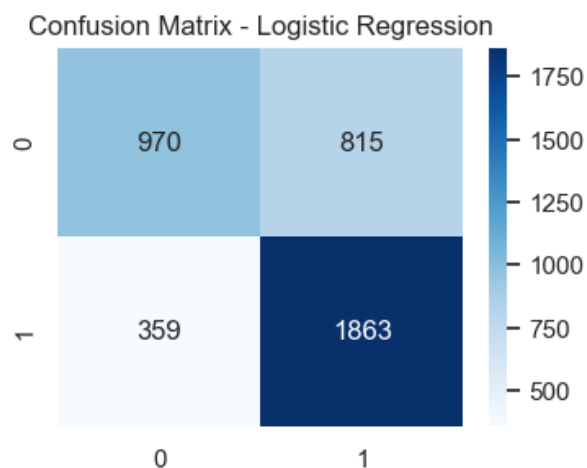


Fig. 2. Confusion matrix for Logistic Regression.

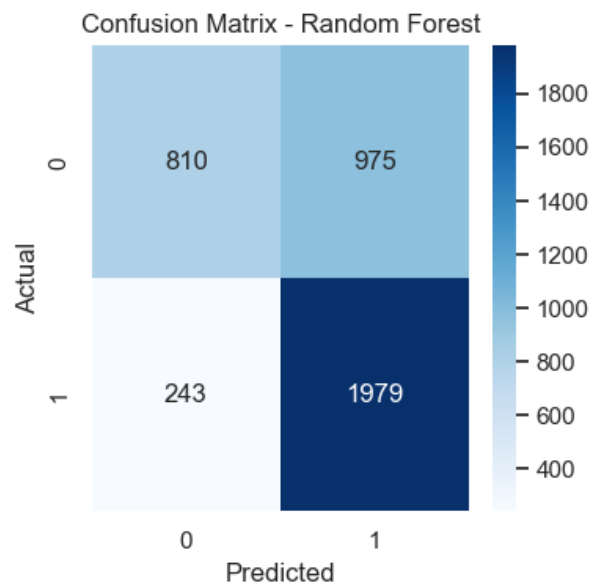


Fig. 3. Confusion matrix for Random Forest.

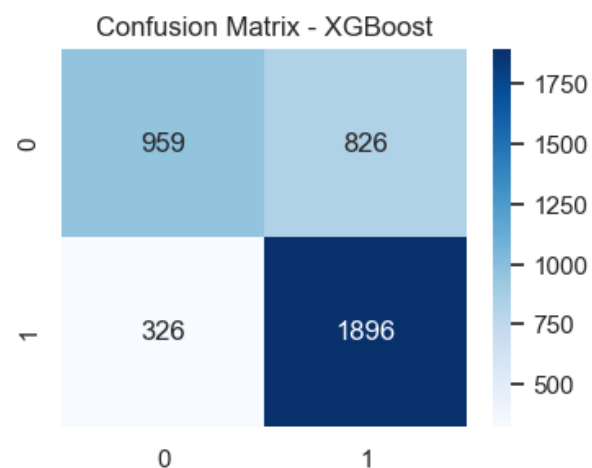


Fig. 4. Confusion matrix for XGBoost.

The confusion matrices reveal that there is a clear biasness towards correctly asking about "Home Wins" (Class 1). All models had a recall of 84 to 89 percent of a home victory. This indicates that the architectures successfully internalized the "Home Court Advantage" found in the training data. The primary error source across the board was "False Positives" in predicting home wins. The models struggled to predict "upsets" where a statistically inferior visiting team defeated a dominant home team. These errors are often tied to factors outside our dataset, such as mid-game injuries or scheduling fatigue (back-to-back games).

D. Elite Team Case Study 2018-19 Toronto Raptors: To determine if the model's predictive power increases when applied to high-performing, consistent teams, we isolated the Toronto Raptors' championship season (where n=89 games).

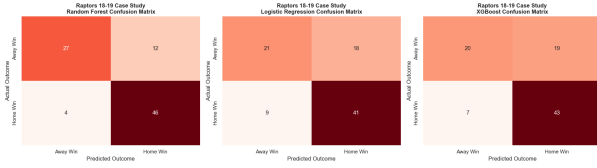


Fig. 5. Model performance on the 2018–19 Toronto Raptors season.

The results for this subset showed a dramatic divergence in model behavior. The Random Forest model experienced a significant performance surge, achieving 82.02% accuracy and an ROC-AUC of 91.92%. This 12.4% jump over the league average suggests that championship teams produce a much "cleaner" statistical signal. While average teams have high win-loss volatility, the 2018-19 Raptors exhibited consistent "elite signatures" in star power and point differentials that the Random Forest was able to identify with near-perfect precision.

E. Predictive Feature Logic Finally, we analyzed which specific data points drove the models final decisions.

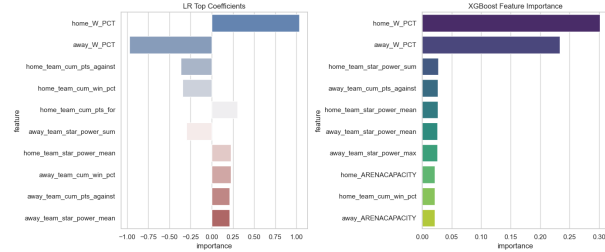


Fig. 6. Top logistic regression coefficients and XGBoost feature importance.

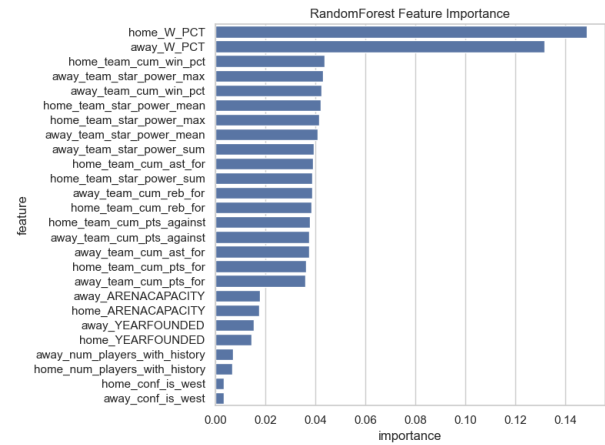


Fig. 7. Random Forest feature importance.

As shown in the importance rankings, Win Percentage and Cumulative Points are the dominant factors. However, the high ranking of the Star Power Sum (the aggregate historical points of the active roster) validates our decision to join player-level datasets with game-level results. This feature allowed the models to "see" roster changes that would be invisible to a model looking only at team names. Furthermore, the inclusion

of Arena Capacity as a top predictor for XGBoost highlights that the environment and "scale" of the home venue serve as a non-linear proxy for the intensity of the home-court advantage.

V. CONCLUSION

This project establishes an efficient method for building multiple machine learning models to predict the outcome of NBA games by utilizing the NBA Datasets Kaggle Repository and extracting key features from league-wide statistics, down to individual player statistics, while accounting for the impact these factors may have on predicting whether the home team will win. The models achieved average test accuracies of 70% and ROC-AUC of 78% across all models when predicting on league-wide data. The results of the case study achieved average test accuracies of 82% and ROC-AUC of 91%. The models do a phenomenal job of predicting the outcome of individual team seasons. Still, they are limited by being slightly inaccurate when predicting outcomes for all teams and by not having access to real-time statistics and contextual factors of the match. Despite that, the methodology established provides a solid structure for more advanced research regarding sports analytics.

VI. FUTURE WORK

Future research that would help the domain of sports analytics and predicting the outcome of NBA games more accurately includes:

Integrating live metrics to support more accurate results across all models Integrating real-time prediction that will update with real-time data while still accounting for historical data Integrating a real-time momentum feature to replicate the effect that momentum has on the game and how it affects the players mentally Hyperparameter tuning to optimize predictive performance Improving model calibration and evaluation methods, such as the Brier score, to make the accuracy of the model reflect the true predicted probabilities Incorporating more defensive and spatial data to track defensive-based metrics, such as off-ball movement and defensive spacing, to account for more context-based factors Researching deep learning in-depth to search for effective ways to build rolling windows that are much larger and accurate

ACKNOWLEDGMENTS

We used ChatGPT for grammar suggestions, minor phrasing adjustments, citation formatting, and assistance in applying the IEEE Overleaf template.

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